FSF3827 Topics in Optimization, 2025
Assignment 3
Due Wednesday December 3 2025 23.59

Instructions for the assignments

- **Collaborations.** It is fine for students to collaborate on solving the assignment. However, every student needs to submit a report and if you have collaborated with someone you need to mention this clearly in the report. If you collaborate, it is essential that you all solve each task and fully understands all parts.
- **Using AI / building upon code you found online.** It is fine to use an AI tool or build upon code that you find online. However, you need to fully understand the code and be able to discuss all parts of the code without help from AI. The teacher will ask questions and make sure you understand what the code does. If you use AI, I recommend you ask it to help with writing parts of the code or subroutines and not full programs.
- The report should have a leading title page where the name, personal number, and e-mail address is clearly stated.
- You are encourage to write the reports in Latex.
- Content
  1. The solutions must be described in enough details that other students (and the teacher) can easily read the report. For example, clearly define variables and sets used in the formulations.
  2. Describe the main parts of the problem formulations. For example, what does the constraints do.
  3. Describe the main parts of the code (what it does and how it works).
- Everyone needs to upload on the Canvas page of the course no later than by the deadline of the assignment:
  - The report as a pdf file.
  - all the relevant code in a zip file.
- You will be graded on a pass/fail scale. If the reports is insufficient, you will get second chance to revise the report.

In this assignment you will work with some classical result on cuts and valid inequalities for mixed-integer programming (MIP), and test some basic concepts. Remember the ability to derive strong valid inequalities is really what makes modern MIP solvers work (without the cuts we would often be left with enumeration).

Tasks

1. Some background: Consider the knapsack set

$$K := \left\{ x \in \{0,1\}^n \ : \ \sum_{j=1}^{n} a_j x_j \leq b \right\}, \qquad a_j > 0,\ b > 0.$$

Definition 1 (Cover)**.** A set $C \subseteq \{1, 2, \ldots, n\}$ is called a *cover* for the knapsack set K if

$$\sum_{j \in C} a_j > b.$$

In other words, the items in C cannot all be chosen simultaneously in any feasible solution of K, so we get the valid inequality (cover-cut)

$$\sum_{j \in C} x_j \leq |C| - 1.$$

Definition 2 (Minimal Cover)**.** A cover C is called *minimal* if no proper subset of C is a cover; that is, for every $j \in C$,

$$\sum_{i \in C \setminus \{j\}} a_i \leq b.$$

Now, we have all we need.

- Assume you have a minimal cover C for the knapsack set K. Show that including one more variable index in set C results in a weaker cover cut.
- Is it also clear that including one more variable in the minimal cover C will result in a weaker cut after lifting in another variable (compared to using the minimal cover)? You can motivate your reasoning by an example.

2. Consider the Multi-Knapsack problem from last home assignment (the text is included at the end of this home assignment).
You don't have to write a general code for this, you can do it "manually" step by step.

- A simple approach for generating cover cuts for this problem is to first solve the LP relaxation. Then based on the variables with non-zero values (at optimum for LP relaxation) you can determine some minimal covers and derive cuts for these. Add all minimal cover cuts you can determine for the relaxation and repeat the procedure (should not be many cuts). Repeat this two times. Do you seem to be closer to an optimal solution?

- Do the same thing, but now also apply lifting to all the cover cuts. Try to obtain as strong cuts as possible by lifting. For more details on lifting see pages 282 - 285 in the book Integer Programming (should be accessible if you are on the KTH network)
`https://link.springer.com/book/10.1007/978-3-319-11008-0`

3. In the last lecture we derived the mixed-integer rounding inequality for a mixed-integer set with a single constraint and two integer variables and a continuous variable. Now consider the general case.

Theorem 1. Consider a mixed integer set defined by a single inequality:

$$S := \{(x, y) \in Z_+^n \times R_+^p : \ ax + gy \leq b\}.$$

Let $f_0 := b - \lfloor b \rfloor$ and $f_j := a_j - \lfloor a_j \rfloor$. Then the inequality

$$\sum_{j=1}^{n} \left(\lfloor a_j \rfloor + \frac{(f_j - f_0)^+}{1 - f_0} \right) x_j + \frac{1}{1 - f_0} \sum_{j: g_j < 0} g_j y_j \leq \lfloor b \rfloor$$

is a valid inequality for convh(S).

- Prove the theorem. You should be able to use a similar approach as we did in the lecture.

4. Read chapters 1, 2, and 4 from the paper "Valid inequalities for mixed integer linear programs" by Gérard Cornuéjols. The paper is in the folder. This is a very good paper on cuts for mixed-integer linear programs, and gives detailed explanations. Read this carefully!
- Chapter 3 is good, but we will come back to this later.

- Let us consider a mixed integer linear set comprising a single equality constraint:

$$S := \left\{ (x, y) \in Z_+^n \times R_+^p : \sum_{j=1}^{n} a_j x_j + \sum_{j=1}^{p} g_j y_j = b \right\}.$$

Let $b = \lfloor b \rfloor + f_0$ where $0 < f_0 < 1$.
Let $a_j = \lfloor a_j \rfloor + f_j$ where $0 \leq f_j < 1$.

Explain how you get the Gomory mixed-integer inequality (GMI)

$$\sum_{f_j \leq f_0} \frac{f_j}{f_0} x_j \ + \sum_{f_j > f_0} \frac{1 - f_j}{1 - f_0} x_j \ + \sum_{g_j > 0} \frac{g_j}{f_0} y_j \ - \sum_{g_j < 0} \frac{g_j}{1 - f_0} y_j \ \geq \ 1.$$

- Explain how "intersection cuts" work and how they are generated (covered in Section 4.9).

5. Gomory cuts (GMI) are typically generated from the simplex tableau. Suppose we have an LP relaxation of our MIP in standard form with the constraints

$$Ax = b.$$

If we solve the relaxation, we can determine the optimal basis for the relaxation and we can write the constraints as

$$Bx_B + Nx_N = b,$$

where B is an invertible matrix. Keep in mind, the optimal solution for the relaxation is $x_N = 0$, and we can write the basic variables as

$$x_B = B^{-1}b - B^{-1}Nx_N. \tag{1}$$

For each row with an integer basic variable and a fractional right-hand side, we can generate a Gomory cut.

Now, consider the Fixed-Charge Network Flow problem described on the last two pages. Write the problem in standard form, solve the relaxation with Gurobi or another solver (to obtain the optimal value of the relaxation, and the current optimal basis), and generate Gomory cuts from all rows with fractional integer variables. Repeat the procedure. Stop if you obtain an integer solution, or if you have added more than 10 Gomory cuts.

You don't need to write a general code for this, you can do it "manually" step by step.

Gurobi is able to solve this problem without branching, just by adding the following cuts

```
Cutting planes:
  Gomory: 3
  Cover: 1
  Implied bound: 3
  Clique: 2
  StrongCG: 1
  Flow cover: 4
  Network: 1
  Relax-and-lift: 2
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But, you may not be able to solve the problem by just adding 10 Gomory cuts. StongCG is a type of Chvátal-Gomory cuts. By now, you should know 4 of the cut types that Gurobi used :)

Good luck!

The multi-Knapsack problem from last home assignment.

In the *Multi-Knapsack Problem (MKP)*, several knapsacks are available, each with its own capacity. An item can be assigned to at most one knapsack, and the goal is to maximize the total profit across all knapsacks. The MKP generalizes the standard KP and frequently appears in various applications.
Consider the MKP problem

$$\begin{aligned}
\text{maximize} \quad & \sum_{j=1}^{2}\sum_{i=1}^{12} p_i x_{ij} \\
\text{subject to} \quad & \sum_{i=1}^{13} w_i x_{ij} \le C_j, \quad j = 1, 2, \\
& \sum_{j=1}^{2} x_{ij} \le 1, \quad i = 1, \ldots, 12, \\
& x_{ij} \in \{0, 1\}, \quad i = 1, \ldots, 13, \ j = 1, 2,
\end{aligned}$$

where

- $m = 2$ is the number of knapsacks, indexed by $j = 1, 2$;
- $n = 12$ is the number of items, indexed by $i = 1, \ldots, 12$;
- p_i is the profit of item i, and w_i is the weight of item i;
- C_j is the capacity of knapsack j;
- $x_{ij} \in \{0, 1\}$ is a binary variable equal to 1 if item i is assigned to knapsack j.

Item i	Profit p_i	Weight w_i	Item i	Profit p_i	Weight w_i
1	22	5	7	50	10
2	18	6	8	15	4
3	35	8	9	21	6
4	14	4	10	19	5
5	60	11	11	24	7
6	12	3	12	16	5

and the capacities of the knapsacks are $C_1 = 22, \quad C_2 = 19$.
I was able to solve the problems by exploring less than 100 nodes (but a bit more can also be correct).

- All the text describing this class of problems have been generated by ChatGPT. Your teacher has only modified some coefficients.

Fixed-Charge Network Flow Model

We consider a classical *fixed-charge network flow* (FCNF) problem, which arises in transportation, logistics, and energy distribution. Flow sent through an arc incurs both a variable cost per unit and a fixed cost if the arc is used at all. A binary variable determines whether the arc is opened.

Network

We are given a directed network with node set

$$N = \{S, A, B, C, D, E\},$$

and arc set

$$A = \{(S, A), (S, B), (S, C), (A, B), (B, C), (A, D), (B, D), (B, E), (C, D), (C, E)\}.$$

Node S is a supply node, nodes D and E are demand nodes, and nodes A, B, C are transshipment nodes. Let b_i denote the net supply at node i. Positive b_i corresponds to demand, and negative b_i to supply. The supplies/demands are:

$$b_S = -40, \qquad b_D = 25, \qquad b_E = 15, \qquad b_A = b_B = b_C = 0.$$

Costs and capacities

Each arc $(i, j) \in A$ has a per-unit variable flow cost c_{ij}, a fixed cost f_{ij} if the arc is used, and a capacity U_{ij}. The data are given in Table 1.

Arc (i, j)	c_{ij}	f_{ij}	U_{ij}
(S, A)	2	30	25
(S, B)	3	18	20
(S, C)	4	10	20
(A, B)	1	8	15
(B, C)	1	6	15
(A, D)	3	20	20
(B, D)	2	16	25
(B, E)	2	14	15
(C, D)	1	12	15
(C, E)	3	5	20

Table 1: Arc costs and capacities for the FCNF instance.

Variables

For each arc $(i,j) \in A$:

$$x_{ij} \geq 0 \quad : \text{flow on arc } (i,j), \qquad y_{ij} \in \{0,1\} \quad : 1 \text{ if arc } (i,j) \text{ is opened.}$$

Mathematical formulation

The objective is to minimize the total fixed charges and variable flow costs, and can be written as

$$\begin{aligned}
\min \quad & 30y_{SA} + 18y_{SB} + 10y_{SC} + 8y_{AB} + 6y_{BC} \\
& + 20y_{AD} + 16y_{BD} + 14y_{BE} + 12y_{CD} + 5y_{CE} \\
& + 2x_{SA} + 3x_{SB} + 4x_{SC} + x_{AB} + x_{BC} \\
& + 3x_{AD} + 2x_{BD} + 2x_{BE} + x_{CD} + 3x_{CE} && (1) \\
\text{s.t.} \quad & x_{SA} + x_{SB} + x_{SC} = 40, && (2) \\
& x_{SA} - x_{AB} - x_{AD} = 0, && (3) \\
& x_{SB} + x_{AB} - x_{BC} - x_{BD} - x_{BE} = 0, && (4) \\
& x_{SC} + x_{BC} - x_{CD} - x_{CE} = 0, && (5) \\
& x_{AD} + x_{BD} + x_{CD} = 25, && (6) \\
& x_{BE} + x_{CE} = 15, && (7) \\
& x_{SA} \leq 25\, y_{SA}, \quad x_{SB} \leq 20\, y_{SB}, \quad x_{SC} \leq 20\, y_{SC}, && (8) \\
& x_{AB} \leq 15\, y_{AB}, \quad x_{BC} \leq 15\, y_{BC}, && (9) \\
& x_{AD} \leq 20\, y_{AD}, \quad x_{BD} \leq 25\, y_{BD}, \quad x_{BE} \leq 15\, y_{BE}, && (10) \\
& x_{CD} \leq 15\, y_{CD}, \quad x_{CE} \leq 20\, y_{CE}, && (11) \\
& x_{ij} \geq 0, \quad y_{ij} \in \{0,1\} \qquad \forall (i,j) \in A. && (12)
\end{aligned}$$

The objective function (1) minimizes the total cost of operating the network. It includes fixed costs for opening arcs through the binary variables y_{ij} and variable costs for sending flow along arcs through the continuous variables x_{ij}.

The flow-balance constraints (2)–(7) ensure conservation of flow at every node. The supply node S satisfies $x_{SA}+x_{SB}+x_{SC} = 40$, each intermediate node has equal inflow and outflow, for example at node A we have $x_{SA}-x_{AB}-x_{AD} = 0$, and each demand node receives exactly its required amount, e.g., at node D the constraint is $x_{AD} + x_{BD} + x_{CD} = 25$.

The capacity-linking constraints (8)–(11) restrict the amount of flow that can use an arc based on its activation. For example, $x_{SA} \leq 25\, y_{SA}$ ensures that arc (S, A) can carry positive flow only if it is opened ($y_{SA} = 1$), and at most 25 units of flow can pass through it.

Finally, the domain constraints (12) enforce that all flow variables satisfy $x_{ij} \geq 0$, and that arc decisions are binary, $y_{ij} \in \{0,1\}$, ensuring each arc is either closed or available for use.